

ALGORITHMIC FAIRNESS 2

Measuring Relative Disadvantage Across Protected Groups

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Module

M515A Ethical Issues for AI

Assignment

Assignment 2: Algorithmic Fairness 2

Dataset

UCI Adult Income Dataset

Models

Logistic Regression, Random Forest, Decision Tree

1. Introduction

This report takes a close look at algorithmic fairness using the UCI Adult Income Dataset — a widely used benchmark drawn from 1994 US Census records. Three machine learning classifiers are trained to predict whether a person earns more than \$50K per year. Once trained, each model is put under the microscope to see how it behaves across two protected attributes: sex and race.

The goal here is purely diagnostic. No corrections are made to the models during training. Instead, the analysis measures baseline discrimination using three complementary lenses: Disparate Treatment, Disparate Impact, and Disparate Mistreatment. Think of these as asking three different questions — are people treated differently?, do different groups receive favourable outcomes at different rates?, and does the model make more mistakes for some groups than others?

Objective

Measure baseline discrimination — no bias mitigation applied

Dataset

UCI Adult Income Dataset (30,162 rows after cleaning)

Target

Income > \$50K vs ≤50K (binary classification)

Sensitive attributes

Sex (Male / Female) | Race (White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other)

Models

Logistic Regression | Random Forest | Decision Tree

Measures

Disparate Treatment | Disparate Impact | Disparate Mistreatment

2. Dataset & Preprocessing

2.1 Dataset Overview

Field	Value
Total Rows (original)	32,561
After Cleaning	30,162
Rows Removed	2,399 (missing: Workclass, Occupation, Country)
Features	14 input features + 1 target
Target Variable	≤50K (0) or >50K (1)
Class Balance	≤50K: 22,654 (75.1%) >50K: 7,508 (24.9%)
Sex — Male	20,380 samples
Sex — Female	9,782 samples
Race — White	25,933 Black: 2,817 Asian: 895 Others: 517

2.2 Data Cleaning

The raw dataset came with a few rough edges that needed tidying before anything meaningful could be done with it:

- Stripped leading and trailing whitespace from all string columns (for example, ' Male' was standardised to 'Male').
- Replaced any 'nan' strings with proper NaN values so they could be handled correctly.
- Dropped all rows with missing values — 2,399 rows in total (Workclass: 1,836, Occupation: 1,843, Country: 583).
- Reset the DataFrame index after dropping rows to keep things clean.

2.3 Data Preprocessing

- The raw Sex and Race columns were saved BEFORE encoding — these are needed later to split the test set into demographic subgroups for fairness analysis.
- Label Encoding was applied to all categorical columns (Workclass, Education, Occupation, Sex, Race, Country, Target, and so on).
- Numeric columns (Age, fnlwgt, Capital_Gain, etc.) were left as-is.
- A stratified 80/20 train-test split (random_state=42) was used to preserve the class distribution.
- StandardScaler was applied to features for Logistic Regression only, since it is sensitive to feature scale.

Train Set

24,129 samples

Test Set

6,033 samples

Positive class (>50K) in test

1,502 samples (24.90%)

Test group sizes — Sex

Male: 4,065 | Female: 1,968

Test group sizes — Race

White: 5,186 | Black: 559 | Asian: 186 | Amer-Indian: 64 | Other: 38

3. Model Training & Accuracy

3.1 Models Used

Logistic Regression:

A linear model that estimates the log-odds of belonging to the positive class. It uses the SAGA solver with `max_iter=2000` and relies on scaled features. It serves as a simple, interpretable baseline — easy to understand but less powerful than ensemble methods.

Random Forest:

An ensemble of 100 decision trees built using bootstrap sampling. It works directly on raw features and is able to capture non-linear interactions between variables — the kind of patterns that a single linear model would miss entirely.

Decision Tree:

A single tree limited to `max_depth=10`. It is the most transparent of the three: every prediction can be traced back through a sequence of feature thresholds. This makes it useful for understanding what the model is actually doing, not just what it outputs.

3.2 Accuracy Results

Model	Accuracy (Raw)	Accuracy (%)
Logistic Regression	0.8321	83.21%
Random Forest	0.8694	86.94%
Decision Tree	0.8467	84.67%

Random Forest edges ahead with 86.94% accuracy, followed by Decision Tree at 84.67% and Logistic Regression at 83.21%. It is worth stressing, however, that none of this accuracy advantage translates into better fairness — as the sections below will show, all three models are equally discriminatory in their predictions.

4. Group Partitioning

After generating predictions, the test set was divided into demographic subgroups using the raw sensitive attribute columns saved before encoding. A helper function called `group_metrics()` handles computing the Positive Prediction Rate, False Positive Rate (FPR), and False Negative Rate (FNR) for each group.

The reference (privileged) groups are:

- Sex: Male — reflecting historically higher income representation in the data.
- Race: White — again reflecting the historical patterns in the 1994 census data.

5. Disparate Treatment

Disparate Treatment asks a straightforward question: does the model predict high income at different rates for different groups, regardless of whether those predictions are correct? It is not about accuracy — it is about whether the model is treating people differently based on who they are.

5.1 Formula

Positive Rate

$P(\hat{y} = 1 \mid \text{group})$

Disparate Treatment Gap (Sex)

$P(\hat{y}=1 \mid \text{Male}) - P(\hat{y}=1 \mid \text{Female})$

Threshold

Gap > 0.05 → UNFAIR | Gap ≤ 0.05 → FAIR

5.2 Results — Sex

Model	Male Rate	Female Rate	Gap (M – F)	Verdict
Logistic Regression	0.2248	0.0412	0.1836	UNFAIR
Random Forest	0.2795	0.1034	0.1761	UNFAIR
Decision Tree	0.2562	0.1089	0.1473	UNFAIR

Every model shows a substantial gap between how often it predicts high income for men compared to women. Logistic Regression has the widest gap (0.1836) — it assigns a high-income prediction to 22.5% of males but barely 4.1% of females. The other two models do slightly better, but still fall far short of the 0.05 threshold. All three models: UNFAIR.

5.3 Results — Race (vs White as reference)

Comparing other racial groups against White as the reference group reveals consistent and significant disparities:

- Logistic Regression: Black gap = -0.1117 (UNFAIR), Amer-Indian-Eskimo = -0.1533 (UNFAIR)
- Random Forest: Black gap = -0.1400 (UNFAIR), Amer-Indian-Eskimo = -0.1965 (UNFAIR)
- Decision Tree: Black gap = -0.1339 (UNFAIR), Amer-Indian-Eskimo = -0.1206 (UNFAIR)
- Asian-Pac-Islander: The only group that comes close to parity — the gap falls below 0.05 for Logistic Regression, though not consistently across models.

6. Disparate Impact

Disparate Impact goes a step further than Disparate Treatment by framing the gap as a ratio rather than a difference. It is the test behind the Four-Fifths Rule used in employment law: if one group receives a positive outcome at less than 80% the rate of the reference group, the system is considered discriminatory.

6.1 Formula

Disparate Impact (DI)

$P(\hat{y}=1 \mid \text{unprivileged}) / P(\hat{y}=1 \mid \text{privileged})$

For Sex

DI = Female Positive Rate / Male Positive Rate

For Race

DI = Minority Positive Rate / White Positive Rate

Four-Fifths Rule

DI < 0.8 → UNFAIR (legally problematic) | DI ≥ 0.8 → FAIR | DI = 1.0 → Perfect parity

6.2 Results — Sex

Model	Male Rate	Female Rate	DI = F/M	Verdict
Logistic Regression	0.2248	0.0412	0.1833	UNFAIR
Random Forest	0.2795	0.1034	0.3700	UNFAIR
Decision Tree	0.2562	0.1089	0.4251	UNFAIR

All three models fail the 80% rule by a wide margin. The most striking case is Logistic Regression, where women receive the positive prediction at roughly 18% the rate of men — less than a quarter of the legal threshold. Random Forest (0.37) and Decision Tree (0.43) are less extreme, but still deeply unfair. No model comes close to the 0.8 benchmark.

6.3 Results — Race (Black vs White)

Model	Group	Group Rate	White Rate	DI	Verdict
LR	Amer-Indian-Eskimo	0.0178	0.1756	0.1014	UNFAIR
LR	Asian-Pac-Islander	0.1534	0.1756	0.8736	FAIR
LR	Black	0.0601	0.1756	0.3422	UNFAIR
LR	Other	0.0547	0.1756	0.3115	UNFAIR
RF	Amer-Indian-Eskimo	0.0328	0.2341	0.1401	UNFAIR
RF	Asian-Pac-Islander	0.2398	0.2341	1.0243	FAIR
RF	Black	0.0912	0.2341	0.3896	UNFAIR
RF	Other	0.0810	0.2341	0.3460	UNFAIR
DT	Amer-Indian-Eskimo	0.0953	0.2198	0.4336	UNFAIR
DT	Asian-Pac-Islander	0.2367	0.2198	1.0769	FAIR
DT	Black	0.0822	0.2198	0.3740	UNFAIR
DT	Other	0.0810	0.2198	0.3685	UNFAIR

The pattern across racial groups is stark. The Black group has a DI ranging from 0.34 to 0.39 across all three models — well below the 0.8 threshold. The Amer-Indian-Eskimo group fares even worse, with DI as low as 0.10 in the Logistic Regression model. Asian-Pac-Islander is the only group to consistently achieve parity, exceeding 0.8 across all models.

7. Disparate Mistreatment

Disparate Mistreatment digs into the types of errors a model makes, not just whether it predicts correctly. Even when overall accuracy looks similar across groups, a model might be systematically missing true high-earners in one group while falsely flagging low-earners in another. This measure catches that kind of asymmetry.

7.1 Formulas

FPR (False Positive Rate)

$FP / (FP + TN)$ — rate of wrongly predicting >50K for someone who earns $\leq 50K$

FNR (False Negative Rate)

$FN / (FN + TP)$ — rate of missing someone who truly earns >50K

ΔFPR

$FPR(\text{Male}) - FPR(\text{Female})$

ΔFNR

FNR(Male) – FNR(Female)

Threshold

$|\Delta\text{FPR}| > 0.05$ OR $|\Delta\text{FNR}| > 0.05 \rightarrow \text{UNFAIR}$

7.2 Results — Sex

Model	FPR M	FPR F	ΔFPR	FNR M	FNR F	ΔFNR	Verdict
Logistic Regression	0.0912	0.0138	0.0774	0.5243	0.7891	-0.2648	UNFAIR
Random Forest	0.1043	0.0277	0.0766	0.3712	0.3854	-0.0142	UNFAIR
Decision Tree	0.0901	0.0339	0.0562	0.4023	0.4011	0.0012	UNFAIR

The results here are particularly striking for Logistic Regression. Its FNR for females is 0.7891 — meaning it misses nearly 79% of women who genuinely earn above \$50K. For men, the equivalent figure is 52%. That is a gap of over 26 percentage points, driven not by the model being deliberately unfair, but by the patterns it has absorbed from historically skewed training data. All three models exceed the 0.05 threshold on at least one error metric.

7.3 Results — Race (Key Findings)

- Logistic Regression: Amer-Indian-Eskimo FNR = 0.8333 vs White FNR = 0.5420 (ΔFNR = +0.2913). The model misses 83% of high-earning individuals in this group.
- Random Forest: Amer-Indian-Eskimo ΔFNR = +0.4725 — the single largest mistreatment gap recorded across all experiments.
- Black group: FNR differences range from 0.127 to 0.208 across models — consistently higher than the White group.
- Asian-Pac-Islander: approaches parity on FPR, but FNR differences remain, especially for Random Forest (ΔFNR = -0.065, UNFAIR).

8. Summary Verdict

The table below brings together all fairness verdicts across every model, attribute, and measure. DT = Disparate Treatment, DI = Disparate Impact, DM = Disparate Mistreatment.

Model	DT Sex	DI Sex	DM Sex	DT Race	DI Race	DM Race
Logistic Regression	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR
Random Forest	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR
Decision Tree	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR	UNFAIR

KEY FINDING: All three models fail every single fairness measure — Disparate Treatment, Disparate Impact, and Disparate Mistreatment — for both sex and race. Not one model passes on any combination. Crucially, the higher accuracy of Random Forest and Decision Tree offers no protection against bias. Being accurate and being fair are not the same thing.

9. Discussion

9.1 Why Does This Bias Exist?

The short answer is that the bias does not originate in the model — it is baked into the data. The UCI Adult Dataset comes from 1994 US Census records, a period when gender and racial pay gaps were substantial and well-documented. Women and minority groups were systematically underrepresented in high-paying roles. When a model is trained on this data, it does not just learn to predict income — it learns to reproduce the inequalities of that era.

This is sometimes called 'historical bias': the model is not behaving unexpectedly, it is doing exactly what a statistical learner should do. The problem is that the patterns it is learning encode social injustice, not neutral truth.

9.2 Why Logistic Regression Has the Worst Disparate Impact

Logistic Regression stands out as the most discriminatory on the DI metric for sex, with a score of 0.18. Because it is a linear model, it assigns a single fixed weight to each feature. If male-associated features correlate strongly with high income in the training data, that relationship gets encoded as a single coefficient and applied uniformly to everyone. Tree-based models, by contrast, can capture non-linear interactions and conditional relationships that partially counteract this — which is why Random Forest and Decision Tree achieve slightly higher DI scores, even though they still fall well short of the legal threshold.

9.3 Limitations

- Correlation, not causation: the disparities shown here reflect statistical patterns in the data, not causal explanations for why they exist.
- Small subgroups: the Amer-Indian-Eskimo group (64 test samples) and Other (38) are small enough that their metrics should be interpreted cautiously — a few more or fewer correct predictions could shift the numbers meaningfully.
- Label Encoding may introduce artificial numerical relationships for tree-based models that do not reflect the true nature of the categorical variables.
- Single train-test split: all results are based on `random_state=42`. Cross-validation would give a more reliable picture of how these models perform in general.
- No intersectional analysis: sex and race are treated as independent attributes. A Black woman, for instance, may face compounded disadvantages that neither dimension captures on its own.

10. Conclusion

This report set out to measure baseline discrimination in three machine learning classifiers trained on the UCI Adult Income Dataset. The findings are unambiguous: every model — Logistic Regression, Random Forest, and Decision Tree — exhibits significant unfairness across both sex and race. There are no exceptions.

Summarising the key results:

- Disparate Treatment: positive prediction rate gaps exceed 0.14 for sex and 0.11 for race (Black vs White) across all models.
- Disparate Impact: DI values for females range from 0.18 to 0.43 — all well below the legal 0.8 threshold. For the Black group, DI ranges from 0.34 to 0.39.
- Disparate Mistreatment: FNR gaps for sex reach 0.26 in Logistic Regression; for Amer-Indian-Eskimo the FNR gap reaches 0.47 in Random Forest.

These results reinforce a point worth taking seriously: standard machine learning models, trained without any fairness constraints on historical data, will reproduce and often amplify the inequalities embedded in that data. Optimising for accuracy is not sufficient. Achieving 86% overall accuracy means nothing if the model systematically misclassifies high-earning women or overlooks entire racial groups.

No bias mitigation was applied in this study. Addressing these disparities would require targeted interventions — whether that means reweighting the training data (pre-processing), modifying the learning objective (in-processing such as adversarial debiasing), or adjusting decision thresholds after the fact (post-processing). These approaches will be explored in Assignment 3.

References

- Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and Machine Learning: Limitations and Opportunities. fairmlbook.org
- Dua, D. & Graff, C. (2019). UCI Machine Learning Repository — Adult Dataset. University of California, Irvine. <https://archive.ics.uci.edu/ml/datasets/adult>
- Feldman, M., Friedler, S., Moeller, J., Scheidegger, C., & Venkatasubramanian, S. (2015). Certifying and Removing Disparate Impact. KDD 2015.